

Lecture 8: Logistic regression, more classification

Max G'Sell
Machine Learning II: 46-927

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Announcements

- ▶ Homework is due Wednesday. There will be a quiz on Friday. Everything is back to the usual schedule.

Today

Today we'll pick back up with logistic regression, and then dive further into classification metrics.

- ▶ Logistic regression
- ▶ Classification:
 - ▶ Imbalanced data
 - ▶ Classification metrics

Where we were

We switched from regression to classification.

Now we observe pairs $(X, Y) \sim \mathcal{P}$, with $X \in \mathbb{R}^p$ and $Y \in \{0, 1\}$ (or sometimes $\{-1, 1\}$).

Based on n observed pairs, we estimate a function $\hat{f}$ that “guesses Y well” for future $(X, Y) \sim \mathcal{P}$.

In a classification problem, we can make $K(K - 1)$ different kinds of mistakes! This leads to two different $K \times K$ matrices reflecting classification performance.

Confusion matrix and loss

Confusion Matrix, which tabulates counts of what actually happened.

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

Loss Matrix, which tabulates how bad each outcome combination is for you.

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	L_{01}
Guess $Y = 0$	L_{10}	0

Confusion matrix: error rates

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

We are interested in many different quantities in the confusion matrix.

The most famous is the misclassification rate: $(FN + FP)/n$

If False positives and False negatives are equally bad, then minimizing the misclassification rate is a good goal.

If we set our loss to have $L_{01} = L_{10} = 1$ and $L_{00} = L_{11} = 0$, minimizing the expected loss will be the same as minimizing the expected misclassification rate.

Statistical decision theory

We showed that the expected misclassification error (or expected prediction error),

$$\text{EPE} = \mathbb{E} \left(\mathcal{L}(Y, \hat{f}(X)) \right) = \mathbb{E} \left(\mathbf{1}_{Y \neq f(X)} \right),$$

is minimized by the ideal rule:

$$f(x) \underset{g \in \{1, \dots, K\}}{\operatorname{argmax}} \mathbb{P}(Y = g | X = x),$$

if we knew $\mathbb{P}(Y = g | X = x)$ for all g . This is called the **Bayes Classifier**, and the error rate it achieves is the **Bayes Rate**.

Building a classifier

Combining this with Bayes' theorem leads to two formulations of an ideal classifier:

$$\begin{aligned}\hat{f}(x) &= \operatorname{argmax}_{j=1,\dots,K} \mathbb{P}(Y = j|X = x) \\ &= \operatorname{argmax}_{j=1,\dots,K} \mathbb{P}(X = x|Y = j) \cdot \pi_j\end{aligned}$$

1. **Discriminative models:** We can try to estimate $\mathbb{P}(Y = j|X = x)$ directly. (Discriminative models)
2. **Generative models:** We could try to estimate the distributions $\mathbb{P}(X = x|Y = j)$ for each class.

Logistic regression

How should we think about modeling $\mathbb{P}(Y = k|X = x)$ directly?

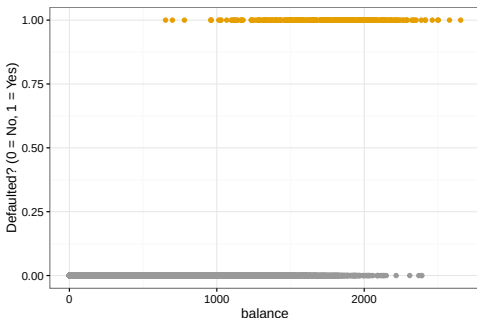
If we only had a few x values, we could directly look at Y conditional on each one:

	Default=No	Default=Yes
Student=No	6850	206
Student=Yes	2817	127

$$P(\text{Default}|\text{Student})$$

$$P(\text{Default}|\text{Not Student})$$

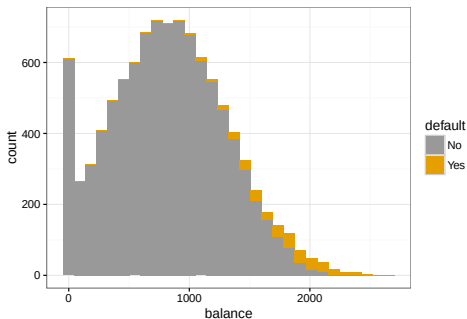
Discriminative models



If we have many values of x , we can't directly look at $P(Y = k|X = x)$ for each x . We need some way to pool information from *similar* points.

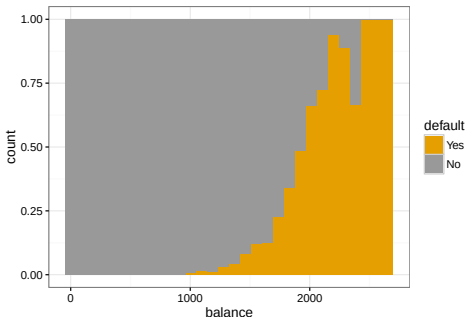
This should remind you of regression.

How should we model $P(Y = 1|X = x)$ for a continuous X ?



A histogram is not quite the right picture for what happens conditional on X , but we can start to see that most individuals do not default, and that large balance seems to be related to default.

This is a conditional density plot. We look at the *probability of default* within each bin.

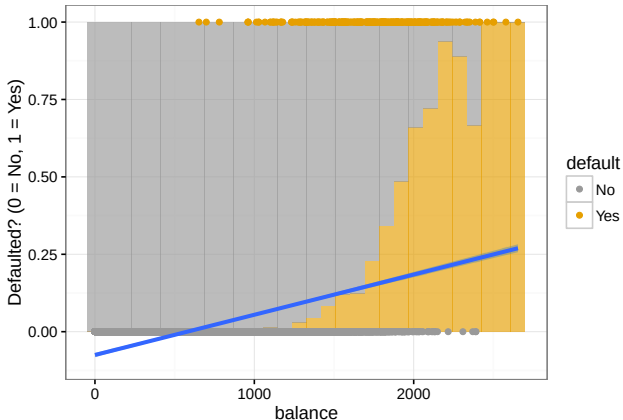


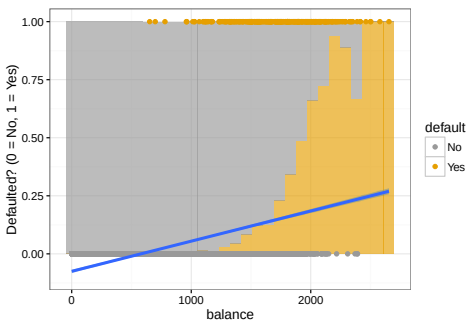
This is a (binned) plot of $P(Y = 1|X = x)$! This is clearly a natural way to think about classification. If I know which bin you are in (X), I can look at how likely you are to default.

How should I build a model of this?

We could try a linear model:

$$P(Y = 1|x = x) = \beta_0 + \beta_1 x$$





Are you happy?

- ▶ Predictions outside $[0, 1]$ don't really make sense.
- ▶ Interpretations of β are strange.
- ▶ Least squares doesn't really make sense for estimation.

Linear regression doesn't work very well for estimating $P(Y = 1|X = x)$, since

- ▶ It doesn't make sense to extrapolate outside of $[0, 1]$.
- ▶ Least squares is an odd way to approximate probabilities.

We still like the idea of forming linear functions of our data, $\beta_0 + x_1\beta_1$ (who doesn't?).

We want a way to squish that linear function back into $[0, 1]$:

$$P(Y = 1|X = x) = \text{squish}(\beta_0 + \beta_1 x_1)$$

Logistic regression

In **logistic regression**, we model

$$\log \left\{ \frac{P(Y = 1|X = x)}{P(Y = 0|X = x)} \right\} = \beta_0 + \beta_1 x + \cdots = \beta^T x$$

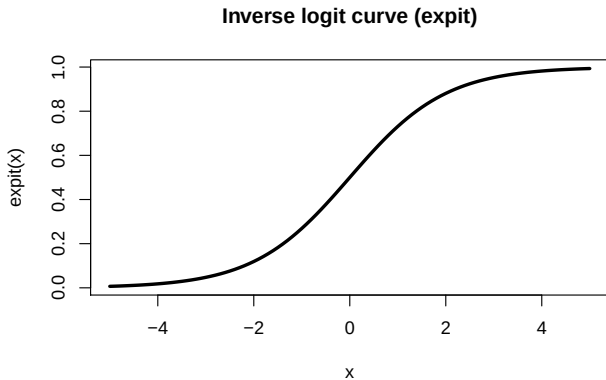
for some unknown $\beta \in \mathbb{R}^p$, which we will estimate directly

Note that $P(Y = 0|X = x) = 1 - P(Y = 1|X = x)$, and

$$\begin{aligned} \log \left(\frac{p}{1-p} \right) &= \beta^T x && \Leftrightarrow \\ &&& \Leftrightarrow \end{aligned}$$

So our model is equivalent to

$$P(Y = 1|X = x) = \frac{\exp(\beta^T x)}{1 + \exp(\beta^T x)}$$



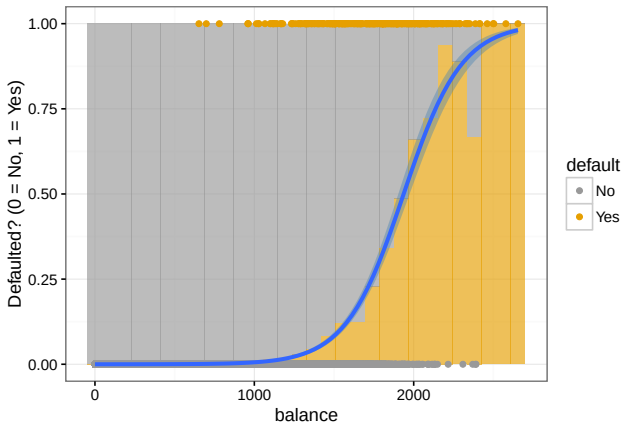
The function

$$\text{logit}^{-1}(z) = \text{expit}(z) = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}}$$

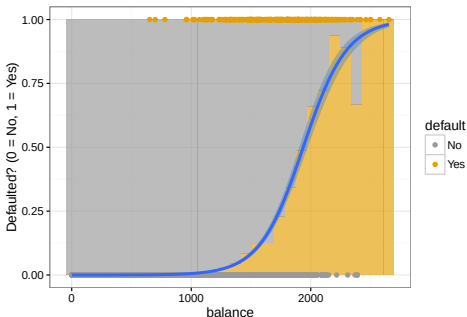
is our desired “squishing” function, transforming real numbers into $[0, 1]$.

Logistic regression and the Default data

The logistic fit gives a much more reasonable estimate of $P(Y = 1|X = x)$!



Logistic regression and the Default data



$$\log \frac{P(\text{Default}|\text{Balance})}{1 - P(\text{Default}|\text{Balance})} = \beta_0 + \beta_1 \cdot \text{Balance}$$

Logistic regression: Estimated probabilities

Once we have estimated $\hat{\beta}_0, \hat{\beta}_1$, we can estimate conditional probabilities:

$$P(Y = 1|X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}$$

For a balance of \$1000 or \$2000,

$$\hat{p}(1000) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1000)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1000)} = 0.00576$$

$$\hat{p}(2000) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 2000)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 2000)} = 0.586$$

Interpretation of logistic regression coefficients

We start to see that the coefficients are *interpretable*.

We have modeled

$$\log \frac{P(Y = 1|X = x)}{P(Y = 0|X = x)} = \beta_0 + \beta_1 x_1$$

The left side, $\log \frac{P(Y=1|X=x)}{P(Y=0|X=x)}$, is called the *log-odds* that $Y = 1$.

This means that the odds that $Y = 1$,

$$\frac{P(Y = 1|X = x)}{P(Y = 0|X = x)} =$$

Increasing x_1 by one unit increases the estimated odds that $Y = 1$ by e^{β_1} .

Classification by logistic regression

Suppose that we fit a logistic regression, estimating $\hat{\beta}$. How do we classify?

Recall that our optimal classifier chooses

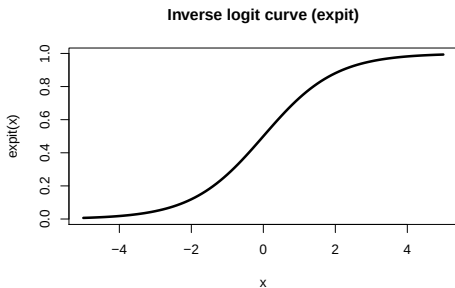
$$\operatorname{argmax}_k P(Y = k | X = x)$$

to minimize 0-1 loss.

Logistic regression gives us an estimate of $P(Y = k | X = x)$! We can just pick the category with the biggest value.

$$\hat{f}(x) = \begin{cases} 1 & \text{if } \frac{\exp(x^T \hat{\beta})}{1 + \exp(x^T \hat{\beta})} > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Classification by logistic regression



Remember that logit and expit are monotonically increasing! This gives us a much simpler rule!

$$\frac{\exp(x^T \hat{\beta})}{1 + \exp(x^T \hat{\beta})} > 0.5 \quad \Leftrightarrow$$

Classification by logistic regression

This gives our final decision rule

$$\hat{f}(x) = \begin{cases} 1 & \text{if } x^T \hat{\beta} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Therefore the **decision boundary** between classes 1 and 0 is the set of all $x \in \mathbb{R}^p$ such that

This is a point in $\mathbb{R}^1$ or a line in $\mathbb{R}^2$.

Estimating logistic regression coefficients

To actually estimate the $\hat{\beta}$, we just use maximum likelihood!

Suppose that we are given an i.i.d. sample (x_i, y_i) , $i = 1, \dots, n$. Here y_i denotes the class $\in \{0, 1\}$ of the i th observation. Then

$$\mathcal{L}(\beta) = \prod_{i=1}^n \text{P}(C = y_i | X = x_i)$$

the likelihood of these n observations, so the log likelihood is

$$\ell(\beta) = \sum_{i=1}^n \log \text{P}(C = y_i | X = x_i)$$

We just plug in our logistic model for these probabilities and optimize.

The log likelihood can be written as

$$\begin{aligned}\ell(\beta) &= \sum_{i=1}^n \log P(C = y_i | X = x_i) \\ &= \\ &= \\ &= \sum_{i=1}^n \left\{ y_i \cdot (\beta^T x_i) - \log (1 + \exp(\beta^T x_i)) \right\}\end{aligned}$$

The coefficients are estimated by **maximizing the likelihood**,

$$\hat{\beta} = \operatorname{argmax}_{\beta \in \mathbb{R}, \beta \in \mathbb{R}^p} \sum_{i=1}^n \left\{ y_i \cdot (\beta^T x_i) - \log (1 + \exp(\beta^T x_i)) \right\}$$

What if I can predict perfectly?

What if I have many variables?

Suppose that the number of variables is large. We would expect the variance of our estimated $\hat{\beta}$ to grow!

What could we do to reduce the variance?

What if we also wanted to pick a sparse, interpretable linear model?

What if we need a more flexible model?

What if we don't really believe that the log odds increase linearly in each of our variables? What if we don't know how to transform our variables?

Classification so far

Focus on binary classification,

- ▶ $Y = 1$ if an event of interest happened
- ▶ $Y = 0$ if it did not happen

Most of our classifiers will give us a guess $\hat{p}(x) = \hat{\mathbb{P}}(Y = 1|X = x)$.
This is an estimate of the true $p(x) = \mathbb{P}(Y = 1|X = x)$.

We saw that to minimize average misclassification error,
 $\mathbb{E}\mathbf{1}_{f(X) \neq Y}$, you should choose $f(x) = \mathbf{1}_{p(x) > 1/2}$.

Let's think about this more carefully.

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

Misclassification rate

Suppose that I tell you my classifier has 97% accuracy. Is this good?

It depends! What if I am predicting lottery wins. The probability of winning may be $1/175000000$.

In that case, guessing "no" has a misclassification rate of 5×10^{-9} ...

We need to consider the *base rate*: the probability of each class given no further information.

Confusion matrix: Example

For the homework problem (to be), $Y = 1$ if an individual responds to marketing. The classification rule $\hat{p}(x) > 0.5$ gives a confusion matrix similar to:

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	2
Guess $Y = 0$	1083	7957

Issues with imbalanced data and misclassification rates

The idea of minimizing the misclassification rate is tempting — no one wants to be wrong!

However, this is often not the real goal of applying a classifier. Losses are usually asymmetric! Sometimes you even care directly about $\hat{p}(x)$!

Furthermore, in settings where the classes are imbalanced, the misclassification rate is very difficult to improve and is often misleading.

Imbalanced data

In many common applications, our classes are imbalanced:

- ▶ Marketing
- ▶ Fraud detection
- ▶ Mortgage default
- ▶ Whether there will be a large jump in price after an event.

Misclassification rate and imbalanced data

We know that we should classify as $Y = 1$ when $P(Y = 1|X = x) > 0.5$ to minimize expected misclassification rate. Suppose that an event only happens 5% of the time in the general population.

By Bayes Theorem,

$$P(Y = 1|X = x) =$$

Therefore, to classify as $Y = 1$, we would need to see an X that is more likely in the $Y = 1$ population than the general population.

Even if there is useful information in the data, it is probably not enough to flip our classifications!

Confusion matrix: error rates

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

Other error rate measures better capture asymmetric concerns when classifying:

- ▶ Sensitivity (Recall, Hit rate):
- ▶ Specificity (True Negative Rate, TNR):

These quantities will allow us to better think about cost trade offs.

Sensitivity and Specificity

Sensitivity: Fraction of points with $Y = 1$ that we find.

- ▶ Want high Sensitivity when False Negatives are more costly than False Positives. Example: Fraud detection.

Specificity: Fraction of points with $Y = 0$ that we avoid.

- ▶ Want high Specificity when False Positives are more expensive than False Negatives. Example: Criminal trials.

We would like both of them to be high, but we need to make tradeoffs as we adjust our threshold.

- ▶ We can flag more cases as fraud to improve sensitivity, but this will decrease our specificity.

Cost/Loss for classification

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	L_{01}
Guess $Y = 0$	L_{10}	0

You will show on your homework that the loss is minimized if you guess $Y = 1$ when

$$\mathbb{P}(Y = 1|X = x) >$$

If it costs \$5 dollars to call someone who will not buy, but it costs \$100 to miss someone who would buy, you should

This suggests that we can trade off costs and error rates by just changing the threshold on $\mathbb{P}(Y = 1|X = x)$!

Changing threshold on $\mathbb{P}(Y = 1|X = x)$

As we change the threshold on $\mathbb{P}(Y = 1|X = x)$, we vary the size and quality of the set we reject.

We can think about $\mathbb{P}(Y = 1|X = x)$ as a function for ranking our observations by likelihood of $Y = 1$.

It is convenient to look at the Sensitivity and Specificity of our classifier as the threshold changes. This corresponds to the ROC (Receiver operating characteristic) curve that you will plot in your homework.

Aside: Classification scores

So far, we've been talking about classification based on estimates $\hat{p}(x) = \mathbb{P}(Y = 1|X = x)$.

In fact, we don't really need probabilities to make classifications. A more general concept is a classification *score*, $\hat{f}(x)$, such that $\hat{f}(x)$ is big for

Any threshold on $\hat{f}(x)$ will yield a confusion matrix, which in turn yields all of the error metrics we have discussed, as well as ROC curves.

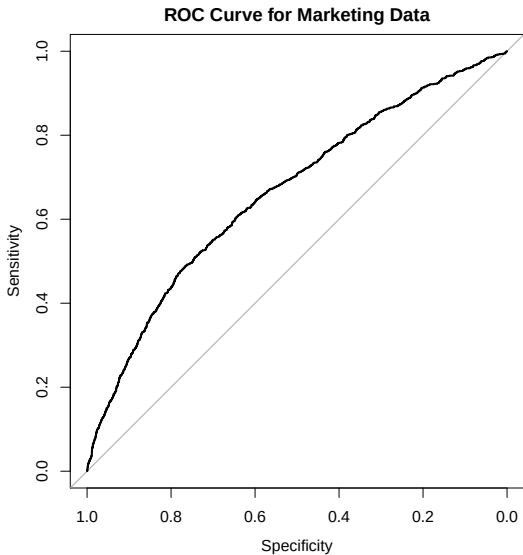
- Question: How do ROC curves change if I monotonically warp my classification score?

Aside: Classification scores

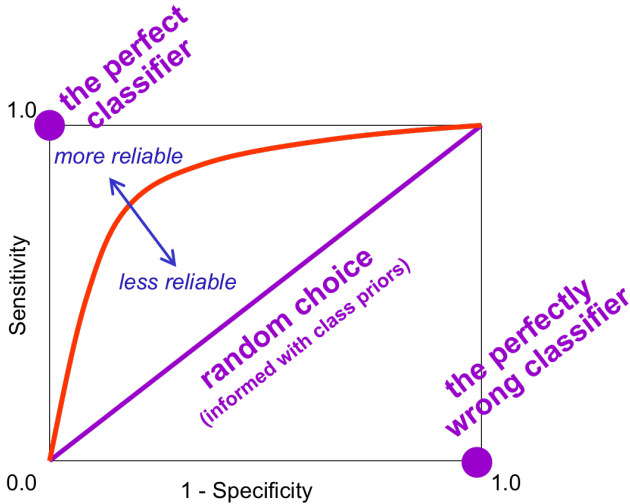
Examples of classification scores that are not probabilities:

It is often useful to think of the distribution of $\hat{f}(x)|Y = k$ for understanding classifier behavior. (Homework)

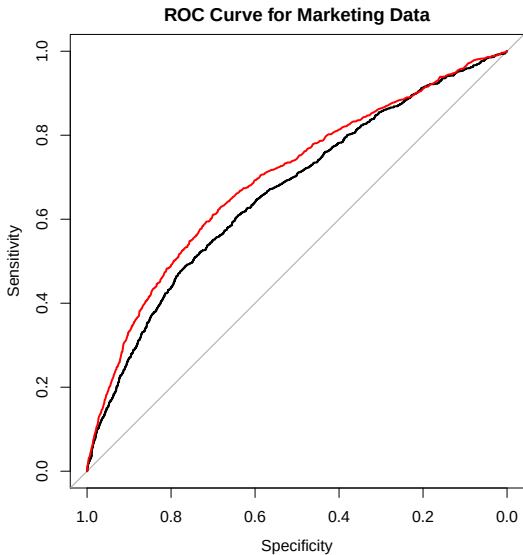
ROC Curve



ROC Curve



ROC Curve (again)



ROC Curve

A generally higher curve tends to be better, though you should keep tradeoffs in mind.

The Area Under the Curve (AUC) is one way to measure this. A higher area tends to be more attractive.

AUC has a nice interpretation:

If you randomly pick two observations from your distribution and order them by $\hat{p}(x)$, the AUC is the probability that your ordering is correct!

In reality though, you are interested in the performance cutoff at a particular point on the curve

Other metrics

Positive predictive value (PPV):

$$\begin{aligned} PPV &= \frac{\# \text{ observations correctly classified as positive}}{\# \text{ observations classified as positive}} \\ &= \frac{TP}{TP + FP} \end{aligned}$$

Suppose that you take a test for a disease. The PPV captures the probability that you actually have the disease when you test positive.

Other metrics

Negative Predictive Value (NPV):

$$\begin{aligned} NPV &= \frac{\# \text{ observations correctly classified as negative}}{\# \text{ observations classified as negative}} \\ &= \frac{TN}{TN + FN} \end{aligned}$$

Suppose that you take a test for a disease. The NPV is the probability that you are not sick, given that the test comes up negative.

Calibration

Is our $\hat{p}(x)$ a good estimate of $p(x) = \mathbb{P}(Y = 1|X = x)$?

Why do we care? We're just going to threshold it...

There is information in knowing $\mathbb{P}(Y = 1|X = x)$.

- ▶ A case that is definitely fraud ($p(x) = 0.99$) may be handled very differently than a case that might be fraud ($p(x) = 0.52$).

Calculating the proper trade-offs for making decisions depends on the probabilities, not just the categories.

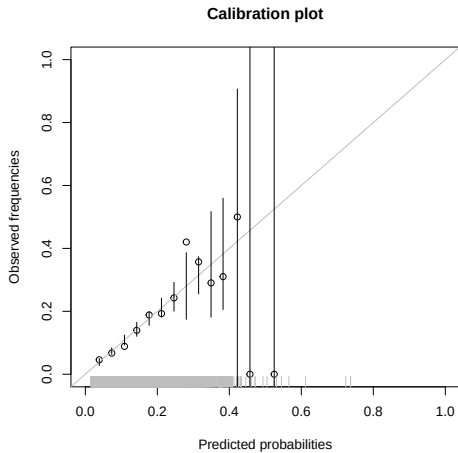
Note: Even if $\hat{p}(x)$ is not a good approximation of $p(x)$, thresholding $\hat{p}(x)$ might still give good guess at categories.

Calibration plots

To check how well calibrated your $\hat{p}(x)$ is, you can make a *calibration plot*:

1. Bin the data according to predicted probability, $\hat{p}(x)$. E.g., $[0, 0.1], (0.1, 0.2], \dots, (0.9, 1]$.
2. For each bin, calculate the proportion of observations with class $Y = 1$
3. Plot these true fractions against the midpoints of the bin predicted probabilities

Calibration plots



Calibration plots

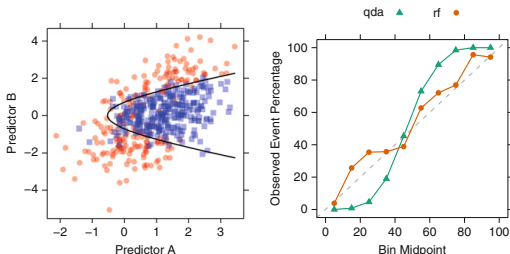


Fig. 11.1: *Left*: A simulated two-class data set with two predictors. The *solid black line* denotes the 50% probability contour. *Right*: A calibration plot of the test set probabilities for random forest and quadratic discriminant analysis models

(From *Applied Predictive Modeling* by Kjell Johnson and Max Kuhn)

If the calibration is off, you can try transforming your $\hat{p}(x)$.

Remark: Model validation and tuning

Suppose that we want to pick between different classification models. Or instead, that we need to choose λ in logistic lasso. How can we do this?